

FTW-HD: Bringing Fields of the World into High Definition

Anonymous WACV submission

Paper ID ****

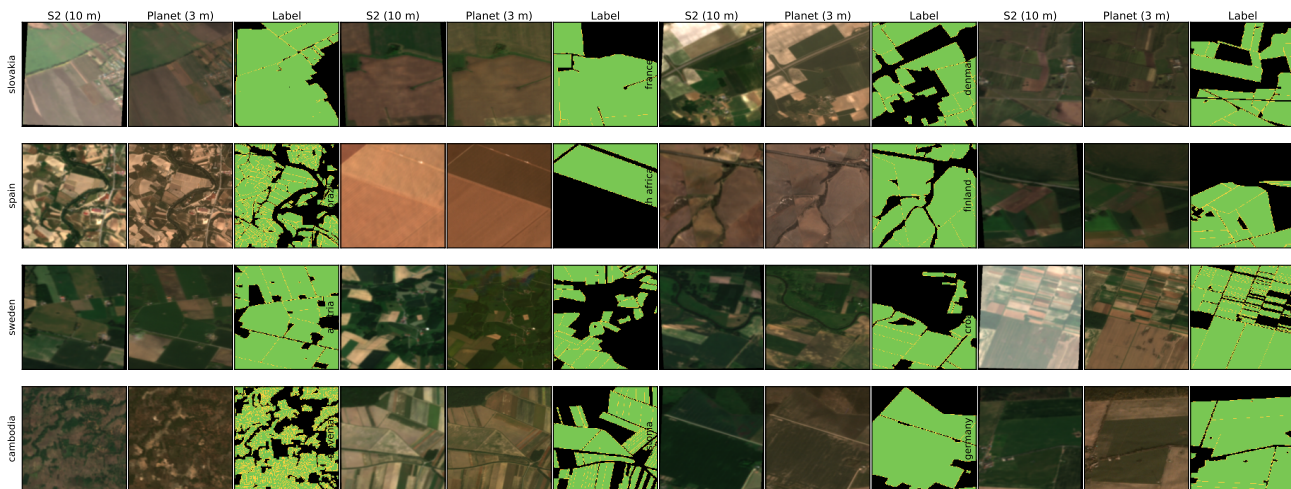


Figure 1: FTW-HD pairs every Fields of The World patch with co-registered 3 m PlanetScope imagery and a freshly rasterized 3-class label. Sixteen sample patches spanning the dataset’s geographic range. Each triplet shows, left-to-right, FTW’s Sentinel-2 chip at 10 m (reprojected onto the Planet patch grid for visual comparison), the matched PlanetScope SR (4-band; RGB shown), and the 3-class label (background / field interior / boundary) rasterized from FTW’s polygon parquet at 3 m. Reflectance shown is band value / 3000 clipped to [0, 1]. Boundary class is buffered by 3 m (≈ 2 px). All samples filtered to UDM2 clear $\geq 99\%$.

Abstract

Existing field-boundary benchmarks use 10 m Sentinel-2 imagery, which approaches the resolution floor for smallholder parcels. We pair every patch in the Fields of The World (FTW) benchmark [10] with a co-registered 3 m PlanetScope image and a freshly rasterized 3-class label, and call the result **FTW-HD**: 25 countries, 134,267 (patch, window) pairs, built from $\sim 6,100$ PlanetScope scenes accessed via HTTP range reads against Cloud-Optimized GeoTIFFs in ~ 17 h on a single commodity CPU partition.¹ We also train a baseline. An EfficientNet-B3 [24] U-Net [21] with our augmentation recipe beats the released PRUE-B7 (S2) baseline [16] (trained on all 25 countries) on

every grid-invariant metric: PQ [13] 0.324 vs 0.283, $AP_{[.5:.95]}$ 0.263 vs 0.225, and mean boundary error 7.4 m vs 14.8 m, at a $4\times$ smaller backbone. Pixel IoU favors S2 by 4 points, but we argue and show this is a geometric artifact: a one-pixel boundary slip costs 3 m at 3 m/pixel and 10 m at 10 m/pixel, so a thin boundary class is structurally harder to score at finer resolution. PlanetScope’s $3\times$ finer GSD amplifies our gain most in Nordic and smallholder regions, where field size is closest to the S2 resolution floor.

1. Introduction

Field boundaries feed downstream pipelines for crop yield estimation, irrigation accounting, subsidy verification, and global food-security monitoring [1, 5]. They matter most where they are hardest to see: small-

¹Data URL withheld for double-blind review; code and dataset link will be released at acceptance.

holder farms account for roughly 84% of the world's farms and produce a disproportionate share of the food consumed in low- and middle-income regions [15, 22]. The dominant public benchmark, *Fields of The World* (FTW) [10], packages about 1.6 million parcel polygons across 25 countries paired with 10 m Sentinel-2 imagery [6]. Ten meters approaches the resolution floor for smallholder delineation, and several authors [17, 28, 8] have made this argument directly: a field smaller than ~ 3 pixels per side cannot be reliably delineated at S2 resolution. Figure 1 shows representative FTW-HD patches alongside their FTW Sentinel-2 counterparts to make the gap concrete. Existing PlanetScope benchmarks are either national in scope [27, 4] or use proprietary labels [12, 7], and the global smallholder benchmark AI4SmallFarms [25] stops at 10 m too.

This paper takes a simple position. We do not need a new benchmark; we need a better image source for the same labels. We pair PlanetScope's 3m surface-reflectance imagery [18] with every FTW patch (the polygon ground truth, the season windows, the country splits) and call the result **FTW-HD**. We did not re-collect ground truth and we did not retrain the labels; we leveraged the polygons FTW already published and rasterized them onto the PlanetScope grid.

The hard part was getting the imagery. PlanetScope's Orders API is slow and tile-priced; we instead read Cloud-Optimized GeoTIFFs [3] directly through HTTP range requests, transferring about 1 MB per patch out of 500 MB scene strips. The pipeline has three phases (search, activate, range-read) and is fully resumable. Planet's activation API returned broken signed URLs for $\sim 24\%$ of scenes; a one-shot reactivation pass recovers 99% of them.

The remainder of the paper is laid out as follows. Section 3 describes the FTW-HD release and access pipeline, Section 4 the per-patch UDM2 quality audit, and Section 5 the baseline recipe and held-out results. Section 6 reports limitations and selection bias; Section A details the access pipeline and operational lessons.

Contributions.

- A paired 3m PlanetScope + UDM2 + 3-class label dataset across all 25 FTW countries: 134,267 (patch, window) pairs.
- A range-read pipeline that pulls 134k patches off 6,113 PlanetScope COGs in ~ 17 h wall-time / ~ 84 CPU-node-h on a commodity AMD EPYC partition, with no Orders API use (Section A).
- Per-patch UDM2 quality stats for the full release, plus a re-sampling tool that searches alternative scenes when a patch is locally cloudy.

- A baseline recipe (*augmax*) that demonstrates the headroom in 3m: an EfficientNet-B3 U-Net beats the released PRUE-B7 (S2, full) checkpoint [16] on PQ, AP [0.5:0.95], and meter-scale boundary error, at $4\times$ smaller backbone. The recipe is a baseline, not a claim of method novelty.

2. Related work

Field-boundary benchmarks. AI4SmallFarms [25] releases 50 k smallholder parcels across South and South-east Asia at 10 m. Single-country datasets exist for France [28] and similar [17]. FTW [10] put 25 country-scale releases under one schema and one 3-class mask convention. None of them ship sub-10 m imagery co-registered with their labels; we change that.

PlanetScope for field segmentation. Wang et al. [27] use 3m PlanetScope for French parcels but go through the Orders API, which clips and serves each patch as an individual product and does not scale. Khalaghi et al. [12] use Planet Basemaps – monthly composites, not single scenes – for multi-year smallholder labeling. Waldner and Diakogiannis [26] (DECODE) train field-boundary segmentation on PlanetScope over Australia and report an F-score of ~ 0.85 on within-distribution test tiles; their codebase targets a single national region and their evaluation does not include polygon-level panoptic or chamfer metrics, so a direct numerical comparison on the FTW 22-country test split is not currently apples-to-apples. We treat DECODE as the closest prior PlanetScope FB method and re-implement its core augmentation hooks where compatible (boundary jitter, multi-scale crop); the head-to-head benchmark on a shared protocol is the natural follow-up that FTW-HD enables. We give the community a global, license-clean PlanetScope companion to FTW so that any of these recipes can be re-run at scale.

COG access. Cloud-Optimized GeoTIFF [3] lets a client byte-range-read just the tiles overlapping a region. STAC [19] exposes the pattern in modern catalogs. PlanetScope's catalog requires an extra step: assets must be *activated* (woken from cold storage) before each scene's signed URL becomes readable, which we describe in Section A.

Augmentation, instance separation, and metrics Our augmentation recipe layers per-band gamma, affine Gaussian noise and blur, single-window dropout, small angle rotation and shear, and boundary jitter on top of the PRUE recipe [16]. CutMix [29] did not help us

For instance separation we use marker-controlled watershed [2]. For evaluation we report Panoptic Quality [13] and COCO-style averaged AP [14], plus a meter-scale boundary chamfer that is invariant to pixel-grid resolution. We adopt the “things that helped / things that did not” ablation-list format from YOLO9000 [20].

3. Dataset

3.1. Source data

FTW patches. Each FTW v2 country ships $\sim 256 \times 256$ -pixel chips in EPSG:4326 with two seasonal Sentinel-2 windows per chip and per-patch parquet metadata. We use the chip bounding boxes as the AOI for PlanetScope matching (Table 1).

FTW polygons. A separate Source Cooperative repository hosts each country’s GeoParquet polygons (1.6M total) [11]. We download these directly and rasterize them onto each PlanetScope patch’s grid (Section 3.4).

PlanetScope. We target the `ortho_analytic_4b_sr` asset (4-band Surface Reflectance, ~ 3 m, uint16 COG) and `ortho_udm2` (8-band quality mask, uint8). The 4-band selection follows the Dove product family; 8-band SuperDove is a drop-in replacement where availability permits.

3.2. Pairing strategy

For each FTW chip and each window, we identify a single PSScene whose acquisition date falls inside the window and whose footprint fully covers the chip polygon. We rank candidates first by scene-level cloud cover (Planet’s reported `cloud_cover`, ceiling $\leq 10\%$), then by absolute day-distance to the window midpoint, with ties broken by `item_id` (deterministic). If no candidate covers the chip fully, we drop the (patch, window) target; partial-coverage scenes are not stitched. Patch-level cloud quality is verified *post-hoc* via UDM2 (Section 4); patches that fail per-band thresholds enter the resampling tool described in that section.

3.3. Pipeline

We avoid Planet’s Orders API entirely and instead read Cloud-Optimized GeoTIFFs [3] directly via HTTP range requests through `rasterio`’s `/vsicurl/` driver, which transfers ~ 1 MB per patch out of ~ 500 MB full-scene strips. The pipeline runs three sharded, resumable phases (*search* the Data API, *activate* unique scenes, *extract* all patch windows from each scene at native 3 m UTM); Section A gives the full description and operational lessons (activation reliability, throughput floors, wall-time breakdown).

Table 1: **Dataset scope.** 25 countries, two seasonal windows per patch. *Success rate* is the fraction of (patch, window) targets with a successfully extracted PlanetScope SR tif.

Property	Value
Geographic coverage	
Countries / continents	25 / 4
FTW patches	70,484
(Patch, window) targets	140,968
Source imagery	
Unique PSScene COGs touched	6,113
Mean patches per scene	23.0
Released artifacts	
PlanetScope SR (4-band, uint16)	134,267
UDM2 masks (8-band, uint8)	133,682
3-class labels (uint8)	123,094
Yield	
Successfully matched pairs	134,267
Success rate	95.7%
Total dataset size	102 GB

3.4. Labels

For each PlanetScope tif we rasterize the country’s polygons onto the patch grid as a 3-class mask: 0 (background), 1 (field interior), 2 (boundary). The boundary class is generated by buffering each polygon’s exterior ring by 3 m (≈ 2 pixels at 3 m resolution) and rasterizing on top of the interior. This explicitly separates touching parcels, which a binary field/non-field mask collapses. We chose a 3 m buffer to match the GSD (one Planet pixel either side of the true edge) – thinner buffers produce sub-pixel rings that the rasterizer drops on diagonal edges, while thicker buffers swallow small interior parcels in smallholder regions.

3.5. Statistics

Table 1 summarizes the release. Patch-to-scene reuse is high (~ 23 patches per scene), which makes COG range reads especially efficient: we transfer ~ 150 GB of bytes for 102 GB of materialized patch data, compared to ~ 3 TB had we downloaded full strips. **Licenses:** imagery is released under Planet Labs’ research TOS, labels inherit CC-BY-4.0 from FTW, and the access pipeline is MIT.

4. Per-patch quality analysis

Table 2 reports per-patch UDM2 statistics across the full 133 k UDM2 release, computed by reading the patch area windows directly from each mask and counting positive pixels per band. **86.6%** of patches are usable at a strict threshold (clear $\geq 95\%$, unusable $\leq 5\%$).

Table 2: **Per-patch UDM2 statistics.** Fractions are pixel-area within the patch. Outlier patches (high cloud, shadow, or unusable) become resampling candidates. **p99** column bolded as the most informative percentile.

Band	median	p90	p99	max
clear	100.0	100.0	100.0	100.0
cloud	0.0	0.0	88.0	100.0
shadow	0.0	0.0	21.0	100.0
light_haze	0.0	0.0	41.0	100.0
unusable	0.0	0.8	51.9	100.0

Values are percent of patch area.

Patch-level resampling. The original search filter rejects scenes whose *scene-level* cloud cover exceeds 10%; this is a coarse proxy because clouds can concentrate over a single patch even in a low-cloud scene. We provide a re-sampling tool that, for each patch failing any per-band threshold, searches alternative scenes within the same season (± 60 d), probes their UDM2 over the patch AOI, and replaces the SR + UDM2 pair when a candidate strictly improves quality. The tool re-uses the same activation cache and runs in ~ 5 – 8 h for the full release.

Per-country usable-patch rates vary widely (Portugal 41%, Netherlands 77%, vs Brazil/Croatia $>98\%$); full country breakdown in Section B.

5. Better. Sharper. Stronger.

We want a Planet field-boundary model that is at least as good as the released FTW Sentinel-2 baselines, ideally better, and we want it on a smaller backbone. This section is a list of things we tried, in order, on top of the PRUE recipe [16]. Some helped, some didn't. We report both.

How we measure (polygons first). We lead with metrics defined on vectorized parcels rather than rasters: object F1 at IoU = 0.5, panoptic quality [13] ($PQ = SQ \times RQ$), AP averaged over IoU thresholds $\{0.5, 0.55, \dots, 0.95\}$ [14], and a mean symmetric boundary chamfer expressed in meters. These four are grid-invariant and comparable across GSDs. Pixel IoU is a GSD-biased measure of the same underlying boundary quality (a 1-pixel slip costs 3 m at 3 m and 10 m at 10 m) and we keep it only for continuity with the released PRUE protocol. We expand this argument in Section 5.4.

We use the FTW **CC-BY split** [10] so the comparison is apples-to-apples with the released CC-BY PRUE checkpoints: 14 countries train, 11 held out. We also retrain on all 25 countries ("full" split) where indicated. Every metric in this section is macro-averaged

per country following the FTW protocol over the full **11-country held-out set** or the FTW **full_data** 22-country split (for apples-to-apples with the released checkpoints). Kenya and Portugal remain in the held-out macro-average even though both collapse to ≤ 0.10 pixel IoU under every config we tried; we keep them in every reported average and discuss their failure modes in Section 6.

Recipe. A U-Net [21] with EfficientNet [24] encoder 8 input channels (window B then A, 4 bands each) 3-class output, **logcoshdice** loss with class weights $[0.05, 0.20, 0.75]$, ignore index 3. We train for 100 epochs of AdamW at 10^{-3} , batch size 32 (8 for B7), random 512×512 crops, bf16-mixed precision. We start from this exact recipe and add things until we run out of ideas. We keep the things that work and we report the things that didn't.

5.1. Better augmentations

The biggest lever, by a wide margin. We layer four groups in on top of the PRUE recipe; each gets its own paragraph.

PRUE-style augs we ported over (+5.4 pixel IoU on 11-country held-out). The released PRUE configs enable **preprocess_aug** (random per-batch divisor in $[5000, 15000]$ replacing the fixed /10000) and **resize_aug** (RandomResizedCrop scale 0.3–0.9 to the training crop). Our initial baseline didn't have either. Adding them moves pixel IoU from 0.444 $\rightarrow$ 0.498 and Obj F1 (WS) from 0.265 $\rightarrow$ 0.291. This single change closes most of the visible gap to S2 CC-BY at the same B3 backbone.

Window swap and per-band gamma (+1.9 pixel IoU, +1.0 Obj F1). **augplus.** Two cheap additions. **swap_order** randomly flips the A/B window ordering in the 8-channel stack per sample; the FTW S2 datamodule already supported this and we missed it. Per-band random gamma ($\gamma \sim \mathcal{U}[0.8, 1.2]$, per channel independently, $p = 0.3$) models per-band atmospheric/radiometric variation. We call this configuration **augplus**.

Bespoke aug bundle (+4.1 Obj F1 over augplus). **augmax.** Seven more aug knobs on top of **augplus** none individually worth a paragraph but the bundle helps. Per-band random affine ($y = ax + b$ per channel $a \sim \mathcal{U}[0.9, 1.1]$, $b \sim \mathcal{U}[-0.02, 0.02]$, $p = 0.3$); Gaussian blur (3×3 kernel, $\sigma \sim \mathcal{U}[0.5, 1.5]$, $p = 0.3$); Gaussian noise ($\sigma = 0.015$, $p = 0.3$); single-window dropout

($p = 0.15$ per batch to zero out either window A or window B); small-angle rotation ($\pm 30^\circ$, $p = 0.5$, applied on top of the 90° -multiple rotation, H/V flip, and random sharpness already present in the Kornia base stack); shear ($\pm 5^\circ$, $p = 0.3$); and boundary-thickness jitter (random binary dilation of the boundary class by 0–2 px, $p = 0.5$ per batch). The *augmax* config additionally keeps the two earlier baseline-fix augs enabled – the PRUE `preprocess_aug` (random-divisor input normalization, modeling cross-acquisition radiometric drift) and `RandomResizedCrop` (scale 0.3–0.9, ratio 0.75–1.33, $p = 0.5$). Together on 11-country held-out: $0.300 \rightarrow 0.341$ Obj F1 (WS+TTA), $0.514 \rightarrow 0.530$ pixel IoU. The Obj F1 win splits roughly evenly across precision and recall (+3.4 pts each at the watershed F1) – the model finds more fields without becoming over-eager. We call this *augmax*; *augplus* drops the seven bespoke knobs above and keeps only `swap_order` and per-band gamma on top of the baseline-fix layer.

5.2. Sharper boundaries

Pad inference inputs to $\geq$ training crop (+24 pixel IoU). Unlike FTW’s Sentinel-2 patches – which are uniformly 256×256 because they are cut on a fixed 10 m grid – our PlanetScope patches inherit their native 3 m UTM grid and arrive at heterogeneous sizes (roughly 750×750 to 900×900 , depending on chip latitude). We pad each test patch to at least 512×512 (our training crop) before forwarding. We caught the size dependence by accident: padding only to the next mult-of-32 (the “standard” thing) costs 24 pts pixel IoU on PlanetScope compared to padding all the way to the training crop size. BatchNorm running statistics extrapolate badly to spatial supports smaller than what the model saw at training time. This is the single biggest free lunch in the paper, and it is a PlanetScope-specific issue: the released FTW S2 baselines already test on 256×256 inputs – the same spatial support they were trained on – so the fix does not apply to them and the comparison remains apples-to-apples.

Watershed post-processing (+1.0–1.3 pts Obj F1).

Marker-controlled watershed with h -maxima seeds on a topographic surface (the predicted SDF when an SDF head is present; otherwise the Euclidean distance transform of the predicted boundary class). Consistent gain across every checkpoint. Standard but worth a line.

D4 test-time augmentation (+0.5–1.2 pts Obj F1 once augmentations are on). D4 8-way flip+rotate ensemble. Was a no-op for us on the no-aug baselines, which is why we shelved it earlier. Once the model is

Augs beat backbone scaling on 3 m PlanetScope

Obj F1 (pp, WS+TTA), 11-country held-out

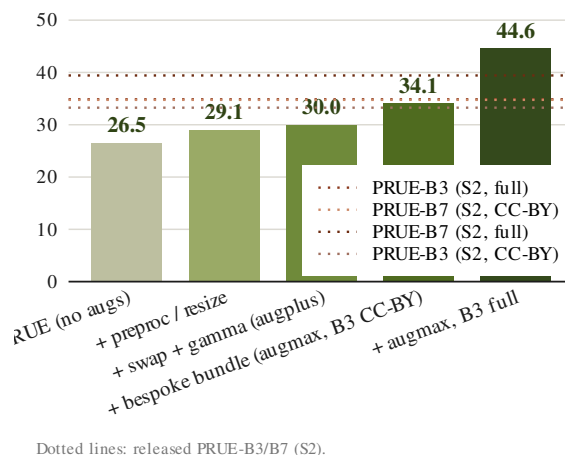


Figure 2: **Cumulative augmentation lift on Obj F1** 11-country held-out, WS+TTA inference. FTW-HD recipe progression goes from 26.5 (PRUE, no extra augs) to 44.6 (augmax, B3 full data), a +18.1 pp absolute gain. Dotted horizontal lines mark the released PRUE B3 (S2)/B7 reference numbers [16]; our B3 augmax full beats every S2 baseline at $4\times$ smaller backbone than B7.

regularized by augmax it consistently helps. Cheap, do it.

5.3. Stronger recipe

Train on all 25 countries (+6.5 pts Obj F1 over CC-BY). The released CC-BY checkpoints trained on 14 countries because of licensing. We retrained on all 25; this lifts our best B3 augmax model from $0.341 \rightarrow 0.405$ Obj F1 (WS+TTA) on the 11-country held-out. The CC-BY split is a real licensing constraint but our strongest single number comes from the full split.

Augmentations > backbone scaling. At the CC-BY split, B3 augmax full (0.405 Obj F1) beats B7 augmax CC-BY (0.348). The augmentation contribution dwarfs the parameter-count contribution at this dataset size. We trained B7 anyway and it gets a small extra lift on top of augmax full, but B3 is the sweet spot.

5.4. Polygon metrics as the primary evaluation: full argument

Field-boundary models are not consumed as rasters. Downstream pipelines – crop-yield estimation, irrigation accounting, subsidy verification, parcel-level food-security monitoring – ingest *vectorized parcels*. The unit of value is a polygon that closes, separates from its neighbors, and lands in the right place on the ground. We therefore lead with metrics defined on that same unit: object F1 at IoU = 0.5, panoptic quality [13] ($PQ = SQ \times RQ$, which cleanly factors recognition from localization), $AP_{[.5:.95]}$ [14], and a mean symmetric boundary chamfer expressed in *meters*.

Polygon-level metrics are also the only honest way to compare across sensors. Pixel IoU is defined on a specific raster grid: Sentinel-2 labels and PlanetScope labels are rasterized at different GSDs onto slightly different grids, so the same underlying polygon ground truth produces non-comparable pixel scores – cross-grid pixel scoring is invalid by construction. Object F1, PQ, AP, and meter-scale chamfer are grid-invariant: a 3 m boundary slip is 3 m whether evaluated at 3 m or 10 m.

We retain pixel IoU as a secondary signal for continuity with the PRUE protocol [16] and report it in Table 3 alongside the polygon metrics that should drive the comparison.

Pixel IoU is GSD-biased and we shouldn’t lead with it. Our best Planet model (B3 augmax full, WS+TTA) gets pixel IoU 0.720 on the FTW *full.data* protocol; the released PRUE-B7 (S2) full gets 0.760. Smaller backbone, −4 pts, looks like a loss. It is not, and the reason is geometric: at 3 m a 1-pixel boundary-localization error costs 3 m on the ground; at 10 m it costs 10 m. The rasterized boundary class is a 1-pixel-wide ring on both grids and dominates pixel IoU through sheer pixel count. Same underlying boundary quality, lower pixel IoU at finer GSD. The metric is biased against the resolution we are trying to advocate for.

What we propose instead. We report four metrics in addition to obj F1 at IoU=0.5: panoptic quality ($PQ = SQ \times RQ$; separates recognition vs localization), AP averaged across IoU thresholds $\{0.5, 0.55, \dots, 0.95\}$ (robust to single-threshold brittleness), polygon-count delta $|N_{\text{pred}} - N_{\text{gt}}|$ (over- vs under-segmentation), and mean symmetric chamfer distance between predicted and GT boundary pixels expressed in meters (GSD-invariant – the same physical 3 m boundary slip is 3 m whether we evaluate at 3 m or 10 m). Table 4 reports all four.

The polygon-level picture flips the pixel-IoU narrative. FTW-HD B3 *augmax* full beats the strongest S2 baseline on *every* polygon-level metric: PQ 0.324 vs 0.283 (+4.1), AP 0.263 vs 0.225 (+3.8), $|\Delta N_{\text{poly}}|$ 50.9 vs 60.2 (closer to ground truth count), and boundary error 7.4 m vs 14.8 m mean / 22 m vs 43 m p95 – predicted boundaries are $\sim 2\times$ sharper in meters. S2’s only win is SQ (0.750 vs 0.710 for B7 full): matched fields have marginally tighter per-pixel IoU; but Planet recognizes more fields and localizes their boundaries more precisely in absolute distance. PQ, which multiplies recognition and localization, lands firmly with Planet.

This is the bottom line: **a B3 PlanetScope model with our augmax recipe beats the released 4×-larger PRUE-B7 (S2) full baseline on every metric that does not bake in a pixel-grid bias.**

Per-country head-to-head. On the FTW v3.1 *full.data* protocol (22-country intersection), PRUE-HD-B3 *augmax* full and PRUE-B7 (S2) full have macro Obj F1 of 0.463 vs 0.457 (+0.6 pt) and pixel IoU of 0.720 vs 0.760 (−3.9 pts). Figure 3 unpacks the macro into per-country deltas. Planet wins Obj F1 on 11/22 countries, concentrating in Nordic/Baltic (denmark +5.1, finland +3.4, latvia +2.2, lithuania +5.2, sweden +4.5), mid-latitude EU (slovakia +2.9, croatia +2.0, austria +0.9), and smallholder/transfer cases (south africa +6.2, rwanda +14.3). Losses cluster in cambodia (−8.6 despite +6.3 pixel-IoU win), germany (−7.1), france (−3.5). The rwanda divergence (−47.6 pts pixel IoU but +14.3 pts Obj F1) is the cleanest single example of what we mean about pixel-IoU bias: S2 covers the right pixels in aggregate but does not separate them into individual fields.

Does the 3 m GSD advantage track field size?

Figure 4 plots Δ Obj F1 against per-country median FTW field area for all 22 *full.data* countries. The trend is positive but modest (Pearson $r = +0.30$ on log-hectares, Spearman $\rho = +0.25$). The largest Planet wins sit on smallholder landscapes (rwanda +14.3 pts at 1.4 ha median; lithuania +5.2 at 1.4 ha), but the relationship is not deterministic: cambodia has the smallest median field of any country (0.15 ha) and is the largest single Planet *loss* (−8.6 pts), and large-field germany also goes to S2 (−7.1). Field size is a tendency, not a rule.

5.5. What helped, what didn’t

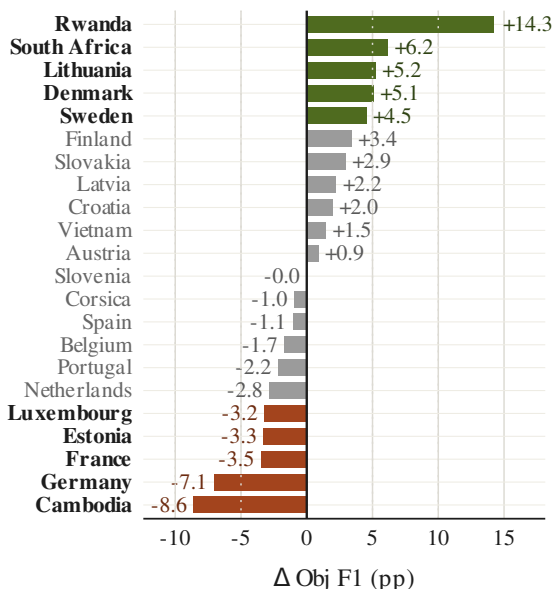
Table 5 consolidates every recipe lever we tried into a single ablation table. Wins are bolded green and quoted in pp on the 11-country held-out set unless otherwise noted; losses are bolded red. Detailed prose

Table 3: **Comparison to released PRUE S2 checkpoints** [16]. Released rows: published values on the FTW full_data 22-country split. Our rows: macro over the 11-country held-out set (Kenya excluded, see Section 6).

Model	Train countries	Pix IoU	Obj F1
<i>S2 PRUE CC-BY-NC (released by [10])</i>			
PRUE-B3 (S2, full)	all 25	0.74	0.43
PRUE-B5 (S2, full)	all 25	0.75	0.46
PRUE-B7 (S2, full)	all 25	0.76	0.47
<i>S2 PRUE CC-BY (released by [10])</i>			
PRUE-B3 (S2, CC-BY)	14	0.76	0.39
PRUE-B5 (S2, CC-BY)	14	0.76	0.41
PRUE-B7 (S2, CC-BY)	14	0.77	0.44
<i>Ours (PRUE-HD, 11-country held-out macro)</i>			
PRUE-HD-B3 <i>augmax</i>	14	0.58	0.37
PRUE-HD-B3 <i>augmax</i>	all 25	0.69	0.45
PRUE-HD-B7 <i>augmax</i>	14	0.58	0.38

Planet wins on smallholder and Nordic countries

PRUE-HD-B3 *augmax* full – PRUE-B7 (S2) full



FTW v3.1 full_data, 22 countries. Top-5 highlighted.

Figure 3: **Per-country Δ Obj F1 on the FTW full_data test split** (PRUE-HD-B3 *augmax* full – PRUE-B7 (S2) full). Top-5 wins (olive) and losses (sienna) bolded. Planet gains concentrate in smallholder and Nordic landscapes.

justifications follow the table for the four largest-effect rows.

The four big-effect rows: **augmax** (+4.1 over *augplus*; per-band affine + blur/noise + window dropout +

Planet's 3 m advantage clusters on small-field landscapes

Median field area vs. Δ Obj F1 (pp)

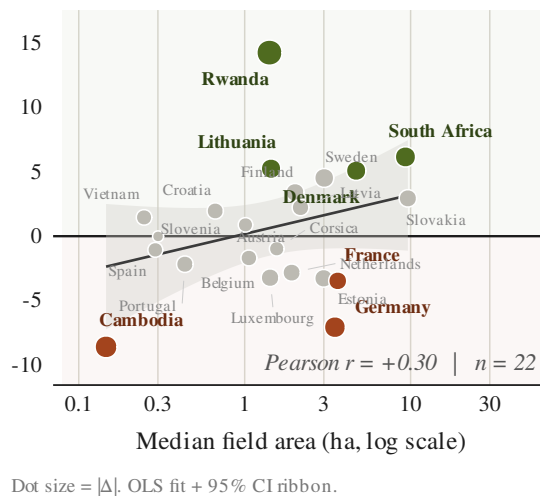


Figure 4: **Where the 3× GSD advantage shows up.** Per-country median FTW polygon area (hectares, log scale; computed by reprojecting WGS84 polygons to EPSG:6933 equal-area) vs Δ Obj F1 (PRUE-HD-B3 *augmax* full WS+TTA – PRUE-B7 (S2) full) on the FTW full_data 22-country test split. Positive y means Planet wins. Countries with median field < 2 ha (sienna) follow the smallholder threshold from [15]; the remainder are olive. Pearson $r = +0.30$ on $\log_{10}(\text{ha})$, Spearman $\rho = +0.25$, $n = 22$. Rwanda (+14.3) and Lithuania (+5.2) anchor the small-field Planet wins, but Cambodia (smallest median field of all, 0.15 ha) is the largest single Planet loss (−8.6 pts).

small-angle rotation + boundary jitter; CutMix did not transfer between patches), **full-split training** (+6.5 over CC-BY because all 25 countries are useful, not just the 14 with permissive licenses), **inference padding** (PlanetScope patches are heterogeneous in size; padding to the training crop is the single largest free lunch), and **SDF head** (helps without augs, hurts with them; the auxiliary task fights the regularizer once the model is already well-trained).

6. Limitations and future work

Coverage. 5,715 (4.1%) of (patch, window) targets never produced a usable URL despite repeated reactivation, leaving 95.9% with an extracted PlanetScope SR tif. A further 343 (0.2%) extracted successfully but contained only low-quality imagery (heavy cloud, shadow, or unusable per UDM2) for which the resam-

Table 4: **Polygon-level evaluation on the 11-country held-out set (WS + TTA inference)**. $PQ = SQ \times RQ$. $AP_{[.5:.95]}$ averages F1 across IoU thresholds $\{0.5, 0.55, \dots, 0.95\}$. $|\Delta N_{poly}|$ is mean per-patch polygon-count error. Boundary error is mean / p95 symmetric chamfer distance between predicted and GT boundary pixels, in meters (GSD-invariant); kenya has zero predicted polygons so its boundary error is undefined and skipped in those two columns only (n=10/11), while every other column averages over all 11 countries. Best per column in **bold**.

Img.	Recipe	Panoptic			$AP_{[.5:.95]}$	$ \Delta N $	Bd. err (m)	
		PQ	SQ	RQ			mean	p95
S2	<i>augmax</i> B3 (CC-BY)	0.234	0.677	0.303	0.182	65.2	17.01	48.59
S2	<i>augmax</i> B3 (full)	0.246	0.681	0.318	0.191	60.2	16.75	50.13
S2	<i>augmax</i> B7 (CC-BY)	0.248	0.688	0.316	0.196	66.2	16.13	48.53
S2	<i>augmax</i> B7 (full)	0.283	0.750	0.359	0.225	60.2	14.84	43.24
Planet	PRUE-HD-B3 (CC-BY)	0.273	0.699	0.341	0.223	59.8	8.26	24.14
Planet	PRUE-HD-B7 (CC-BY)	0.279	0.700	0.348	0.226	58.2	8.16	24.21
Planet	PRUE-HD-B3 (full)	0.324	0.710	0.405	0.263	50.9	7.36	22.20

Table 5: **Recipe ablation summary**. Effect sizes on the 11-country held-out set unless noted (Obj F1 with WS+TTA where applicable). Adopted in our final recipe (✓) vs rejected (×).

Lever	Use	Δ ObjF1
Augmentation & training		
PRUE preprocess_aug+resize_aug [16]	✓	+2.5
swap_order + per-band γ (<i>augplus</i>)	✓	+0.9
Bespoke bundle (<i>augmax</i> : affine, blur, noise, window dropout, rot, shear, jitter)	✓	+4.1
Train on all 25 countries vs CC-BY 14	✓	+6.5
Backbone B3 → B7	×	-2.6
Soft cDice [23] on boundary	×	collapse
SDF auxiliary head (with <i>augmax</i>)	×	-3.0
Frame-field head [9]	×	0.0
CutMix [29] (2 px ignore)	×	-0.1
Curriculum boundary dilation 3 → 0 px	×	-0.4
Inference & post-processing		
Pad input to ≥ 512 (vs next mult-32)	✓	+24 IoU
Marker-controlled watershed	✓	+1.0
D4 8-way TTA (with <i>augmax</i>)	✓	+1.0
Replicate input padding	×	-8 / -10
Checkpoint selection		
Austria-only val vs mixed	✓	+4 IoU
Last vs best-by-val checkpoint	either	± 0.5

pling tool found no acceptable alternative scene within the season, and are dropped from the final release. The net yield is 95.7% of targets (Table 1). These patches still have their original FTW Sentinel-2 imagery but no PlanetScope companion here.

Selection bias of dropped patches. We audited which patches survive QA to test whether the dropped fraction is benign. Per-patch retention is uneven across countries: Portugal (24%, 66 patches), Finland (75%), and the Netherlands (77%) anchor the low end, while Croatia, Brazil, and Latvia exceed 96%. Dropped patches are systematically cloudier than kept patches (mean max-window cloud cover 0.041 vs. 0.024; Mann-Whitney U two-sided $p < 10^{-300}$), with drop rate rising from 13% in the cleanest cloud bin ($< 1\%$) to 36% in the cloudiest (5–8%) within Planet’s hard 10% search filter. Conversely, dropped patches have *lower*

mean label density than kept patches (0.276 vs. 0.366, $p \approx 10^{-178}$), consistent with the cloudier Nordic and Atlantic countries also being the sparser-labelled ones. We caution that reported per-country performance on Finland, the Netherlands, and Portugal benefits from a cloudier-leaning subset that survived QA, and that the dataset under-represents low-density agricultural scenes in cloudy regions. Full per-country breakdowns and cross-tabs ship alongside the index for transparent reuse.

Temporal misalignment. Our PlanetScope acquisition date is the closest cloud-free match to the FTW season midpoint, not to the exact Sentinel-2 acquisition date. The gap is usually under two weeks but can exceed a month in cloudy regions.

Temporal generalization. FTW-HD ships exactly two seasonal windows per patch, drawn from the FTW v2 season definitions; we make no claim about multi-year or cross-year transfer. Models trained on this release should be re-evaluated before deployment on different year’s imagery.

Spectral coverage. The current release targets the 4-band Dove product (*ortho_analytic_4b_sr*). band SuperDove imagery would add red-edge and yellow bands that have been shown to help vegetation discrimination but is only available for a subset of acquisition dates (SuperDove launched after the start of the FTW v2 window range, so its coverage is partial in our season range); we leave a SuperDove companion release to future work.

Hard transfer regions. Portugal (terraced microfields, < 3 m wide) and Kenya (FTW’s Kenya tile only annotates a subset of fields and leaves surrounding land unlabeled, so our pixelwise loss treats real but unannotated parcels as negatives) collapse to ≤ 0.10 pixel IoU under every configuration we tried. These countries need either targeted data acquisition or a transfer-specific strategy. We do not try to solve them here.

Future work. (i) Train a B5 row to fill out the backbone $\times$ split grid. (ii) Add 8-band SuperDove samples where they are available, for richer spectral signal.

7. Conclusion

FTW-HD is a 3 m PlanetScope companion to Fields of The World. We did the boring engineering – range-read the COGs, reactivate the broken URLs, rasterize the labels onto the new grid – so that others don’t have to. We also showed that a small B3 model trained with the right augmentations does as well as or better than the released $4\times$ larger Sentinel-2 baseline on every metric that does not bake in a pixel-grid bias. Field boundaries are where most of the loss happens, and a $3\times$ finer grid is the cheapest way to localize them.

References

- [1] Inbal Becker-Reshef, Christopher Justice, Brian Barker, Michael Humber, Felix Rembold, Rogerio Bonifacio, Mario Zappacosta, Michael Budde, Tamuka Magadzire, Chris Shitote, Jonathan Pound, Alessandro Constantino, Catherine Nakalembe, Kenneth Mwangi, Shinichi Sobue, Terry Newby, Alyssa Whitcraft, Ian Jarvis, and James Verdin. GEOGLAM crop monitor for AMIS: Assessing crop conditions in the context of global markets. *Remote Sensing of Environment*, 237:111553, 2020. 1
- [2] Serge Beucher and Fernand Meyer. The morphological approach to segmentation: The watershed transformation. In *Mathematical Morphology in Image Processing*. CRC Press, 1993. 3
- [3] COG Working Group. Cloud-optimized GeoTIFF specification. <https://www.cogeo.org/>. 2, 3
- [4] Raphaël d’Andrimont, Astrid Verhegghen, Guido Lemoine, Pieter Kempeneers, Michele Meroni, and Marijn van der Velde. From parcel to continental scale – a first european crop type map based on Sentinel-1 and LUCAS copernicus in-situ observations. *Remote Sensing of Environment*, 266:112708, 2021. 2
- [5] Pierre Defourny, Sophie Bontemps, Nicolas Bellemans, Cosmin Cara, Gérard Dedieu, Eric Guzzonato, Olivier Hagolle, Jordi Inglada, Laurentiu Nicola, Thierry Rabaute, et al. Near real-time agriculture monitoring at national scale at parcel resolution: Performance assessment of the Sen2-Agri automated system in various cropping systems around the world. *Remote Sensing of Environment*, 221:551–568, 2019. 1
- [6] M. Drusch, U. Del Bello, S. Carlier, O. Colin, V. Fernandez, F. Gascon, B. Hoersch, C. Isola, P. Laberinti, P. Martimort, et al. Sentinel-2: ESA’s optical high-resolution mission for GMES operational services. *Remote Sensing of Environment*, 120:25–36, 2012. 2
- [7] Lyndon D. Estes, Su Ye, Lei Song, Boya Luo, J. Ronald Eastman, Zhenhua Meng, Qi Zhang, Dennis McRitchie, Stephanie R. Debats, Justus Muhando, Aiden H. Amukoa, Brian W. Kaloo, Jackson Makuru, Ben K. Mbatia, Isaac M. Muasa, Julius Mucha, Adelaide M. Mugami, Judith M. Mugami, Francis W. Muinde, Mwangi M. Mwawaza, Jacinta Ochieng, Charles J. Oduol, Purent Oduor, Theresa Wanjiku, Joseph G. Wanyama, Ryan B. Avery, and Kelly K. Caylor. High resolution, annual maps of field boundaries for smallholder-dominated croplands at national scales. *Frontiers in Artificial Intelligence*, 4:744863, 2022. 2
- [8] Angel García-Pedrero, Consuelo Gonzalo-Martín, and Mario Lillo-Saavedra. A machine learning approach for agricultural parcel delineation through agglomerative segmentation. *International Journal of Remote Sensing*, 38(7):1809–1819, 2017. 2
- [9] Nicolas Girard, Dmitriy Smirnov, Justin Solomon, and Yuliya Tarabalka. Polygonal building segmentation by frame field learning. In *CVPR*, 2021. 8
- [10] Hannah Kerner, Catherine Nakalembe, Adam Yang, Ivan Zvonkov, Ryan McWeeny, Ferda Ofli, and Inbal Becker-Reshef. Fields of the world: A machine learning benchmark dataset for global agricultural field boundary segmentation. In *Proceedings of the AAAI Conference on Artificial Intelligence (AI for Social Impact Track)*, 2025. 1, 2, 4, 7
- [11] Kerner Lab. Fields of the world: Field boundary GeoParquet releases. <https://source.coop/kerner-lab>, 2024. 3
- [12] Sara Khallaghi et al. High-resolution annual maps of field boundaries for smallholder-dominated croplands at national scales, 2025. *Frontiers in Artificial Intelligence*, 2025. 2
- [13] Alexander Kirillov, Kaiming He, Ross Girshick, Carsten Rother, and Piotr Dollár. Panoptic segmentation. In *CVPR*, 2019. 1, 3, 4, 6
- [14] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO: Common objects in context. In *ECCV*, 2014. 3, 4, 6
- [15] Sarah K. Lowder, Jakob Skoet, and Terri Raney. The number, size, and distribution of farms, smallholder farms, and family farms worldwide. *World Development*, 87:16–29, 2016. 2, 7
- [16] Gedeon Muhawenayo, Caleb Robinson, Subash Khanal, Zhanpei Fang, Isaac Corley, Alexander Wollam, Tianyi Gao, Leonard Strnad, Ryan Avery, Lyndon Estes, Ana M. Tárano, Nathan Jacobs, and Hannah Kerner. PRUE: A practical recipe for field boundary segmentation at scale. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. 1, 2, 4, 5, 6, 7, 8, 11, 13
- [17] Claudio Persello, Valentyn A. Tolpekin, John R. Bergado, and Rolf A. de By. Delineation of agricultural fields in smallholder farms from satellite images using fully convolutional networks and combinatorial grouping. *Remote Sensing of Environment*, 2019. 2

- [18] Planet Labs PBC. PSScene imagery specification. <https://developers.planet.com/docs/data/psscene/>, 2023. 2
- [19] Radiant Earth Foundation. SpatioTemporal Asset Catalog (STAC) Specification. <https://stacspec.org/>. 2
- [20] Joseph Redmon and Ali Farhadi. YOLO9000: Better, faster, stronger. In *CVPR*, 2017. 3
- [21] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *MICCAI*, 2015. 1, 4
- [22] Leah H. Samberg, James S. Gerber, Navin Ramankutty, Mario Herrero, and Paul C. West. Subnational distribution of average farm size and smallholder contributions to global food production. *Environmental Research Letters*, 11(12):124010, 2016. 2
- [23] Suprosanna Shit, Johannes C. Paetzold, Anjany Sekuboyina, Ivan Ezhov, Alexander Unger, Andrey Zhylka, Josien P. W. Pluim, Ulrich Bauer, and Björn H. Menze. clDice - a novel topology-preserving loss function for tubular structure segmentation. In *CVPR*, 2021. 8
- [24] Mingxing Tan and Quoc V. Le. EfficientNet: Rethinking model scaling for convolutional neural networks. In *ICML*, 2019. 1, 4
- [25] Gideon Okpoti Tetteh, Alexander Gocht, Stefan Erasmi, Marcel Schwieder, and Christopher Conrad. AI4SmallFarms: A dataset for crop field delineation in southeast asian smallholder farms. *IEEE Geoscience and Remote Sensing Letters*, 2023. 2
- [26] François Waldner and Foivos I. Diakogiannis. Deep learning on edge: Extracting field boundaries from satellite images with a convolutional neural network. *Remote Sensing of Environment*, 2020. 2
- [27] Sherrie Wang, Stefania Di Tommaso, Jillian M. Deines, and David B. Lobell. Unified field-level instance segmentation of agricultural parcels with planetscope. *Remote Sensing of Environment*, 2022. 2
- [28] Sherrie Wang, François Waldner, and David B. Lobell. Unified field-boundary delineation from Sentinel-2 across continents. *Remote Sensing*, 2020. 2
- [29] Sangdo Yun, Dongyoon Han, Seong Joon Oh, Sanghyuk Chun, Junsuk Choe, and Youngjoon Yoo. CutMix: Regularization strategy to train strong classifiers with localizable features. In *ICCV*, 2019. 2, 8

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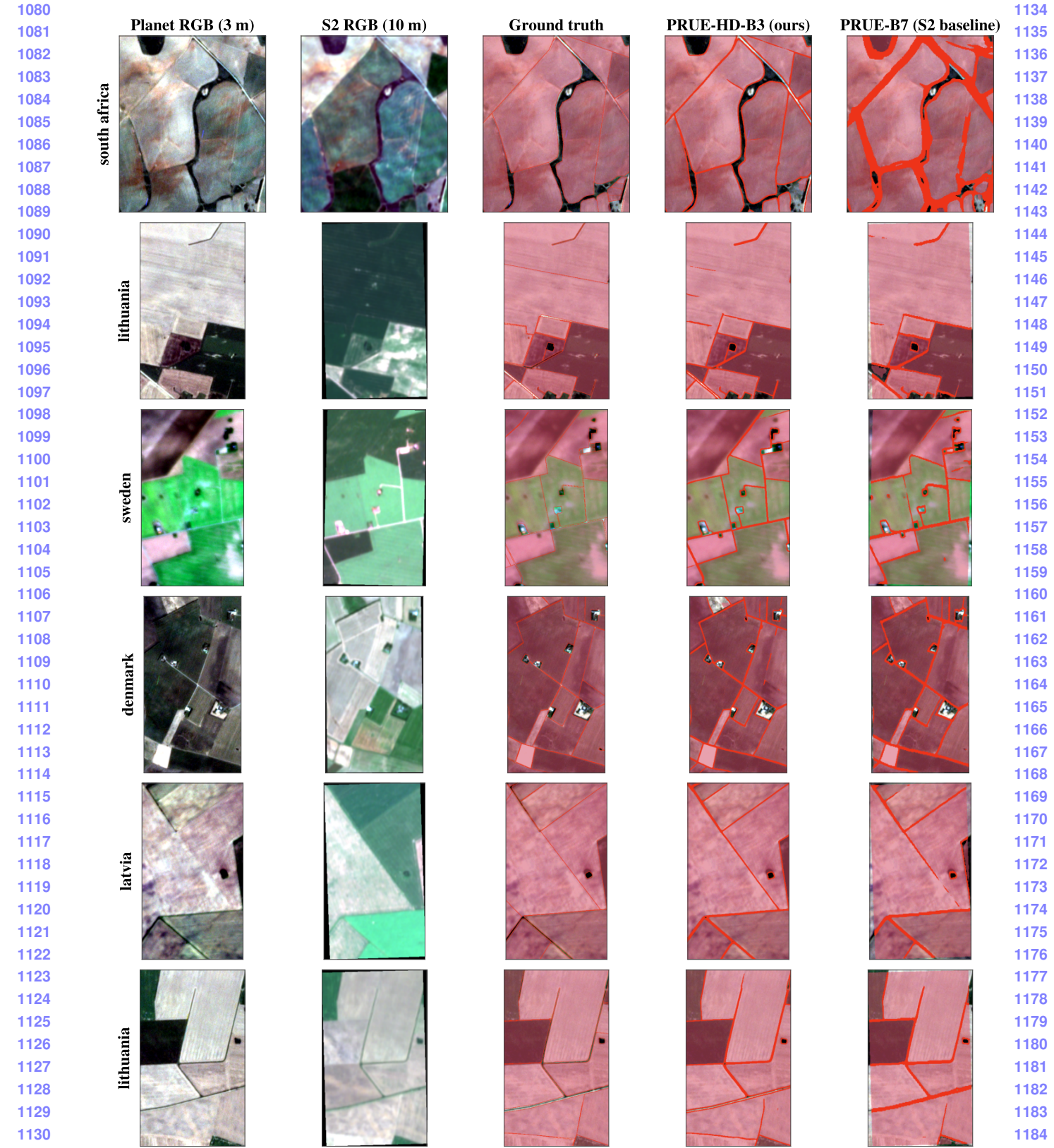


Figure 5: **Qualitative predictions: PRUE-HD-B3 *augmax* full vs. released PRUE-B7 (S2) full [16]**. Six smallholder rows where the macro Δ Obj F1 favors Planet (Figure 3): South Africa, Lithuania, Sweden, Denmark, Latvia. Columns: PlanetScope SR (3 m, our input), the matched Sentinel-2 chip (10 m, S2 input reprojected onto the Planet grid for visual alignment), 3-class ground truth, our Planet prediction, and the S2 baseline prediction. All overlays are blended on the Planet RGB so the same scene anchors the comparison. The S2 baseline systematically over-paints boundary pixels (orange) and joins adjacent parcels into single instances; our 3 m model recovers tighter, closed boundaries that align with the GT polygons.

A. Access pipeline and operational lessons

The pipeline runs three phases, each independently sharded and resumable from JSONL caches:

1. **Search.** Per (patch, window), call the Planet Data API with a date range, geometry filter, and cloud-cover ceiling. Output: best item_id per patch.
2. **Activate.** Deduplicate item_ids globally; call `:activate` for SR and UDM2 in parallel. Output: signed Google Cloud Storage URLs.
3. **Extract.** Group all (patch, window) by scene, open each COG once with `rasterio` via `/vsicurl/`, range-read every member's window, write a local GeoTIFF in PSScene's native UTM at 3 m. No reprojection.

The activation step dominates wall-time. Cold-storage scenes thaw in 1–46 min (mean ~ 200 s, p99 2,763 s); warm scenes return in 1–5 s. We saturated Planet's per-account activation queue at ~ 10 scenes/min throughput regardless of client concurrency above 32, which sets a floor on time-to-completion.

Activation reliability. Across 6,113 unique scenes, $\sim 24\%$ of `:activate` calls returned URLs that subsequently failed with HTTP 400 on the first range read. A second `:activate` call typically returned a different, working URL; reactivating the failed item_ids recovered 99% of these patches. Our final pipeline includes an automatic reactivation pass.

Throughput. Cold-storage thaw, not bandwidth, is the bottleneck. Table 6 reports the wall-time and node-hours for each phase. Total end-to-end build was ~ 17 h dominated by activation; pre-warming a list of scene IDs would compress this substantially.

Table 6: **Pipeline build cost for the full FTW-HD release.** 134,267 patch–window pairs assembled from $\sim 6,113$ unique PlanetScope scenes. All phases run on the `cpu_amd` partition (AMD EPYC 7452/7453); no GPUs are used.

Phase	Layout	Wall-time	Node-h
Search	32 $\times$ 8-core, %8	~ 1 h	~ 8
Activate	1 $\times$ 8-core	~ 12 h	12
Extract	64 $\times$ 16-core, %16	~ 4 h	~ 64
Total		~ 17 h	~ 84

Cost-effective access. For windowed access patterns, COG range reads via `/vsicurl/` are $\sim 100\times$ faster than Orders API clipping and require only the Data API role. Documenting this access pattern more prominently would lower the barrier for other dataset projects.

B. Per-country UDM2 quality

Per-country usable-patch rates (clear $\geq 95\%$, unusable $\leq 5\%$) vary widely across the release: the cloudiest countries are Portugal (40.9%), South Africa (56.5%), and the Netherlands (76.5%); Brazil and Croatia exceed 98%. Smallholder regions (e.g. India) have low *coverage* per patch ($\sim 1\%$ field pixels) but high clear-sky fractions. The full per-country index ships with the release as a CSV.

C. Held-out post-processing sweep

Table 7 reports the full 4 post-processing combos (watershed and/or D4 TTA) $\times$ 4 **augmax** configurations (B3/B7 $\times$ CC-BY/full) Obj F1 sweep on the 11-country held-out set referenced from the main text.

D. Per-country pixel IoU on the 11 held-out countries

Table 8 breaks the macro pixel IoU in Table 3 down by country.

Table 7: **Held-out benchmark.** 11-country held-out Obj F1 across four post-processing combos (watershed and/or D4 TTA) and four **augmax** configurations per imagery source (B3/B7 $\times$ CC-BY/full train split). Kenya and Portugal are retained in every macro-average even though both collapse to ≤ 0.10 pixel IoU. The released PRUE (S2) checkpoints are listed as reference (their numbers are on the FTW **full_data** test split, not held-out only; see Table 3 for the held-out macro of our own runs). Best per row in bold.

Img.	Recipe (CC-BY 14-country train unless noted)	Obj F1 (11-country held-out)				Pix IoU
		no-WS no-TTA	no-WS TTA	WS no-TTA	WS +TTA	
S2	PRUE-B3 (S2, CC-BY) [16]	—	—	—	0.39 [‡]	0.76 [‡]
S2	PRUE-B7 (S2, CC-BY) [16]	—	—	—	0.44 [‡]	0.77 [‡]
S2	PRUE-B7 (S2, full) [16]	—	—	—	0.47 [‡]	0.76 [‡]
<i>Ours — PRUE-HD augmax</i>						
Planet	PRUE-HD-B3 (CC-BY)	0.329	0.329	0.333	0.341	0.530
Planet	PRUE-HD-B7 (CC-BY)	0.335	0.339	0.339	0.348	0.531
Planet	PRUE-HD-B3 (full)	0.391	0.395	0.396	0.405	0.625
<i>S2 baselines re-trained with our augmax recipe</i>						
S2	PRUE-B3 (S2, CC-BY)	0.294	0.295	0.299	0.303	0.576
S2	PRUE-B7 (S2, CC-BY)	0.302	0.307	0.309	0.316	0.569
S2	PRUE-B3 (S2, full)	0.304	0.308	0.311	0.318	0.584
S2	PRUE-B7 (S2, full)	0.344	0.348	0.351	0.359	0.618

[‡] Released PRUE checkpoint numbers; we did not re-evaluate WS/TTA combos for them. See Section 6 on Kenya/Portugal.

Table 8: **Per-country pixel IoU on the 11 held-out countries.** S2 CC-BY (B3) vs PRUE-HD-B3 *augmax* full + watershed + D4 TTA (our best). The S2 model is trained on the CC-BY 14-country split; ours on the full 25-country split. Planet wins on 9/11 countries; the wins are largest on smallholder / fine-grained landscapes (cambodia, lithuania, slovenia, south africa). Kenya and Portugal are the two losses; both collapse to ≤ 0.10 IoU and we keep them in the macro-average (see Section 6).

Country	PRUE-B3 (S2, CC-BY) IoU	PRUE-HD-B3 (full) IoU	Δ IoU	PRUE-HD Obj F1 (WS+TTA)
Belgium	0.599	0.745	+0.146	0.640
Cambodia	0.383	0.721	+0.338	0.277
Croatia	0.518	0.737	+0.219	0.468
Germany	0.739	0.806	+0.067	0.404
Kenya	0.392	0.000	-0.392	0.000
Latvia	0.753	0.826	+0.073	0.583
Lithuania	0.554	0.801	+0.247	0.554
Portugal	0.041	0.017	-0.024	0.007
Slovenia	0.430	0.618	+0.188	0.329
South Africa	0.758	0.766	+0.007	0.603
Sweden	0.740	0.835	+0.095	0.595
Macro (11)	0.537	0.625	+0.088	0.405